Uniform and scalable light-sheets generated by extended focusing

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Abstract: Light-sheet fluorescence microscopy (LSFM) affords highly parallelized 3D imaging with optical sectioning capability and minimal light exposure. However, using Gaussian beams for light-sheet generation results in a trade-off between beam waist thickness and the area over which the beam can approximate a light-sheet. Here, we present a novel form of LSFM that uses incoherent extended focusing to produce divergence free light-sheets with near diffraction-limited resolution and uniform intensity distribution along the propagation direction. We demonstrate the imaging performance of the new technique by volumetric imaging of beads, collagen fibers, and melanoma cancer cells with sub-cellular resolution.

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OCIS codes: (110.0110) Imaging systems; (170.0170) Medical optics and biotechnology; (180.0180) Microscopy; (000.1430) Biology and medicine; (110.6880) Three-dimensional image acquisition.

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1. Introduction

Imaging of dynamic three-dimensional sub-cellular processes requires high spatiotemporal resolution combined with low phototoxicity, and preferably, it should be performed within a native or biomimetic context (*e.g.*, *in vivo,* or within a reconstituted extracellular matrix [1]). Widefield epifluorescence microscopy offers rapid data acquisition, limited only by signal flux and camera frame-rate, but fails to provide three-dimensional optical sectioning. In contrast, confocal microscopy rejects out-of-focus fluorescence and enables true 3D imaging, but laser scanning substantially decreases the temporal resolution, degrades signal to noise, and is often accompanied by nonlinear photodamage mechanisms.

Light-sheet fluorescence microscopy (LSFM) provides an alternative imaging technology that overcomes many of the challenges faced by widefield and confocal microscopy [2, 3]. In its simplest form, LSFM consists of excitation and detection objectives placed orthogonally. The sample is typically illuminated from a low-NA excitation objective with a long depth-of-focus line-shaped (*e.g.*, shaped with a cylindrical lens [2]) or laterally scanned Gaussian beam [4]. The goal is to confine the excitation power to the focal plane of the detection objective, thereby avoiding out-of-focus fluorescence and lowering the overall light exposure in 3D imaging. By scanning the excitation beam in the Z-direction synchronously with the focal plane of the detection objective, volumetric data sets with near beam-waist-limited axial resolution are obtained.

LSFM is particularly useful for imaging applications that require large field of views at moderate spatial resolution (on the order of a couple of microns). For example, LSFM has provided significant insight into zebrafish embryogenesis and neuronal dynamics [5–9]. Sub-cellular imaging, however, requires sub-micron, preferably diffraction-limited, axial resolution, where beam divergence significantly limits the field of view. In an attempt to increase the field of view, Bessel-Gauss beams (a Bessel beam of finite length) have been adopted [10, 11]. However, Bessel-Gauss beams have concentric side lobes that significantly degrade the axial confinement of illumination, necessitating optical-sectioning structured illumination (OS-SIM) [11], super-resolution SIM (SR-SIM) [12], or confocal detection [8, 13] to reject out-of-focus light. With 2-photon excitation, these side lobe structures can be suppressed significantly, which results in a pencil of light that can be used to generate a nearly ideal sheet of light [11]. Nevertheless, 2-photon excitation has limited multicolor capability because it requires spectrally resolvable fluorochromes with overlapping 2-photon absorption cross-sections, emission wavelength tuning of the Ti-sapphire laser, or complex and expensive optical parametric oscillators. Furthermore, because photodamage scales non-linearly with excitation intensity, 2-photon excitation generally increases cellular phototoxicity and photobleaching compared to 1-photon imaging [12, 14].

To overcome the limitations of 1-photon based LSFM, we propose the use of extended focusing to produce uniform, divergence free light-sheets with near diffraction-limited resolution. In this paper, we generate an extended focus by incoherently superimposing a

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diffraction-limited laser focus in the axial direction. We first numerically simulate the properties of light-sheets generated by incoherent extended focusing and compare the results to light-sheets generated by Bessel beams.

As proof of principle, we implemented the extended focus scheme with an acousto-optical focus tunable lens that sinusoidally sweeps a focus along the optical axis at hundreds of kHz, generating a time averaged extended focus. The resulting pencil of light is scanned laterally with a galvanometric mirror to synthesize a digital light-sheet. To further reduce out-of-focus blur, we synchronized the rolling shutter of a sCMOS camera with the sweep of the pencil beam, generating a virtual confocal slit aperture [13, 15]. We validate imaging performance *in silico*, on sub-diffraction fluorescent beads, on live metastatic melanoma cells expressing fluorescently labeled actin, and on a fluorescently labeled collagen matrix [16].

2. Materials and methods

2.1 Simulations

All numerical simulations were performed with Matlab (MathWorks) running on a Linux server equipped with 168 CPU nodes and 384 Gb of RAM.

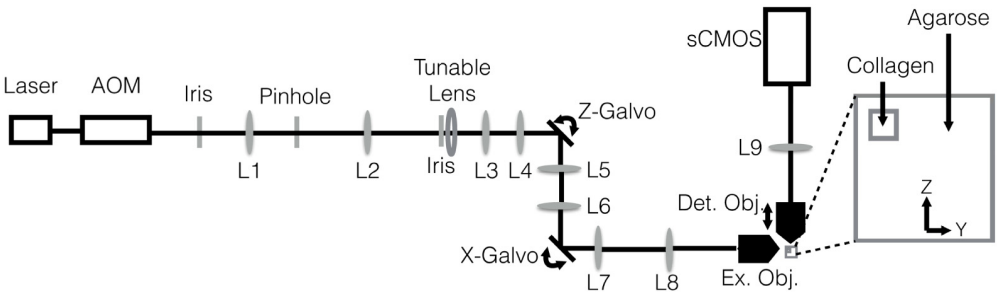

Fig. 1. Microscope configuration.

2.2 Microscope configuration

Figure 1 shows a schematic representation of our imaging system. A 488 nm continuous wave laser (Coherent, Sapphire, 500 mW) was shuttered with an acousto-optic modulator (IntraAction Corp., AOM-402AF1), and the first-order diffracted beam was spatially isolated. Laser intensity control was performed by altering the drive voltage of the AOM. The laser beam was spatially filtered (focal lengths of L1 = 50 mm, L2 = 200 mm) through a 30-micron pinhole, magnified to a diameter of 3.4 mm (e^{-2}), and directed into an acoustically tunable lens (TAG Optics Inc., TL25.Beta.B.VIS). The lens was resonantly driven at ~191 kHz (small differences in temperatures lead to shifts in the resonance up to 1 kHz, which is continuously adjusted for by the driver of the lens), yielding a useful aperture of up to 4 mm in diameter. Per specifications, the optical power can be changed over 8 diopters (+ –250 mm). The TAG lens was imaged 1:1 (L3 = 50 mm, L4 = 50 mm) onto the Z-galvo (Cambridge Technology, 6210HSM40B), relayed with 1.5x magnification (L5 = 50 mm, L6 = 75 mm) onto the X-galvo (Cambridge Technology, 6210HSM40B) and finally relayed with a telecentric F-theta lens (L7 = 60 mm, Sill Optics, S4LFT0061/065) and an infinity-corrected tube lens (L8 = 200 mm, ThorLabs, ITL200) onto the back focal plane (BFP) of a long working distance water-dipping objective (Nikon 16x, 0.8 NA).

With an effective aperture of 4 mm for the TAG lens and the further downstream magnification it is possible to completely fill the BFP of the illumination objective. In practice, we experienced aberrations (presumably stemming from the TAG lens and some of the used optical components) that restricted us to using only NA = 0.6, which corresponds to a diameter of 15 mm at the BFP of the illumination objective. The effective axial focus change can be estimated with Eq. (1) [17], where FL3 is the focal length of lens L3 and M is

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the magnification from the sample plane to the intermediate image plane inside the telescope formed by the two lenses L3 and L4, and η is the refractive index of the immersion medium. FL_{TAGMIN} and FL_{TAGMAX} are the minimal and maximal focal lengths for the TAG lens, respectively. This results in an axial focus sweep over 162 microns.

$$\boldsymbol{\delta Y} = \frac{\eta * FL3}{M^2 * FL_{TAGMIN}} - \frac{\eta * FL3}{M^2 * FL_{TAGMAX}} \quad (1)$$

For detection, a long working distance water-dipping objective (Nikon 40x, 0.8 NA), placed orthogonally to the excitation objective, is synchronously scanned with the Z-galvo using a long travel range objective piezo (Mad-City Labs, Nano-F450). The fluorescence emission is filtered with a 488 nm long-pass filter (488 nm EdgeBasic, Semrock), and imaged by an infinity-corrected tube lens (ThorLabs, ITL200) onto the active area of a scientific complementary metal-oxide-semiconducter (sCMOS) camera (Hamamatsu, Flash 4.0). With the TAG lens operating at 191 kHz and a 20 ms camera integration period (512 by 512 region of interest) each pixel row is illuminated ~15x, which is sufficient to create a uniform time-averaged line illumination. Synchronization of the laser shuttering, image acquisition, galvo sweeps, and piezo scanning, was controlled with a field-programmable gate array (National Instruments, PCIe-7852R) executing custom LabView software (National Instruments) written by Coleman Technologies. All images were saved as OME-tiffs, and image analysis was performed using ImageJ (National Instruments of Health) or Matlab.

2.3 Mammalian cell culture and labeling

MV3 melanoma cells were a kind gift from Dr. Peter Friedl, and were cultured in Dulbecco's modified essential medium (DMEM) supplemented with 10% fetal bovine serum (FBS) and penicillin-streptomycin. An N-terminal fusion of mNeonGreen [18] and actin with a 7 amino-acid linker (SGLRSRA, Allele Technologies) was integrated into the genome of the MV3 cells using lentivirus (Clontech, pLVX-IRES-Puro), and puromycin was used to eliminate non-fluorescent cells.

2.4 Sample preparation

Samples were prepared in a custom teflon mold that enabled casting of an ultra-pure 2% agarose cube (Life Technologies) onto an anodized aluminum holder. To measure the point-spread function of the microscope, 200 nm beads (Polysciences, Inc) were directly embedded within the agarose gel. For biological imaging, a stainless steel square peg excluded a small volume of agarose while it solidified. Removal of the peg created a void where trypsinized MV3 cells could be placed within a polymerizing 2% collagen gel (Advanced BioMatrix, PureCol Bovine Collagen, Type I). To directly image collagen, FITC-labeled collagen (Sigma-Aldrich) was mixed with 2% unlabeled bovine collagen, pH-neutralized, incubated at 4 degrees Celsius for 1 week, and immediately prior to imaging, polymerized at 37 degrees Celsius. For imaging, the void containing polymerized collagen was positioned at the intersection of the optical axes of the excitation and detection objectives. The precise position of the gel was adjusted by translating the aluminum holder with a manual XYZ stage (ThorLabs, PT3).

3. Results

3.1 Simulations

To simulate the extended focus, a uniform and circular electric field with a vacuum wavelength of 488 nm is propagated through an ideal lens of NA = 0.6, and is forward propagated plane-by-plane to yield a three-dimensional distribution of the electric field. Hence we ignore polarization effects and apply the scalar theory of light. The squared modulus of the electric field provided the 3D intensity distribution of the laser focus. The lateral full width at half maximum (FWHM) of the laser focus was 389 nm, which is in good agreement with Abbe's resolution limit. To obtain the incoherent extended focus, the simulated laser focus was convolved with a line 50- and 100-microns in length along the

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optical axis. To capture almost all out-of-focus blur and to minimize wrap-around effects by the convolution, the initial laser focus was computed in a large simulation volume of 204 x 204 x 200 microns (2206 x 2006 x 541 voxels). Figures 2(a) and 2(b) show the intensity distribution of the incoherently extended focus in the axial and lateral dimensions, respectively. The lateral FWHM of the beam is 465 nm, which increased by 19.5% compared to the initial laser focus.

The three-dimensional Bessel-Gauss beam was modeled in reciprocal space: voxels on a thin annulus (NA range 0.57 to 0.58) on a sphere were set to one whereas all other elements in a three-dimensional matrix were set to zero. A fast Fourier transform (FFT) of the matrix yielded the three-dimensional electric field in real space and the intensity distribution was obtained by taking the squared modulus of each voxel. Matlab and cluster performance set an upper limit for the size of the matrix for which a three-dimensional FFT could be computed. The size of the matrix was chosen to be M x M x P = 2300 x 2300 x 1060, where M is the array size along the lateral dimensions and P is the array size in the propagation direction of the beam. The radius of the sphere, which is equal to the wavenumber of light, was set to 580 voxels. Hence the reciprocal voxel size is given by $\delta k = \eta/(\lambda\ r)$, where η is the refractive index of water (1.333), λ is the vacuum wavelength of light and r is the radius of the sphere. The real space voxel size can be computed as $\delta xy = (M \cdot \delta k)^{-1} = 92.5$ nm and $\delta z = (P \cdot \delta k)^{-1} = 183.5$ nm, which are sufficient to Nyquist sample the beam in the respective dimensions.

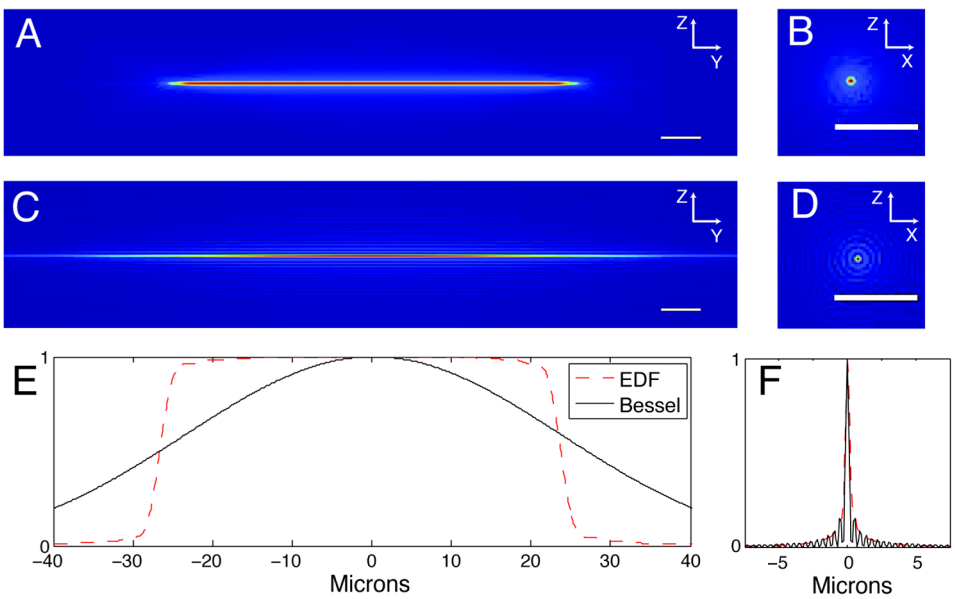

Fig. 2. Simulated intensity distributions. Scale bars are 5 microns. (A) Extended depth-of-focus Gaussian beam in the propagation direction. (B) Cross-section of extended depth-of-focus Gaussian beam. (C) Intensity distribution of a finite Bessel beam in the propagation direction. (D) Cross-section of Bessel beam. (E) Profile plots of the simulated beams in the propagation and (F) lateral dimensions.

Figures 2(c) and 2(d) show the cross-sections of the Bessel-Gauss beam in the axial and lateral directions, respectively. The lateral FWHM of the Bessel-Gauss beam amounted to 288 nm. As the annulus in reciprocal space has a finite width, the propagation length of the Bessel-Gauss beam is also finite and exhibits a Gaussian intensity distribution [Fig. 2(e)] with a FWHM of 55 microns. The extended focus beam, convolved with a 50 micron long line, has approximately the same FWHM, but the axial beam profile looks analogous to a top hat, providing even illumination over ~50 microns and decaying rapidly at its ends. With the Bessel-Gauss beam, a similar evenness can only be achieved over a narrow region around its peak (~10 microns).

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The aforementioned simulations provide the intensity profile for a single beam. However, to generate a digital sheet of light, the beam must be rapidly scanned in the X-direction. To obtain the intensity distribution of a scanned beam, the cross-sectional intensity distribution [as shown in Figs. 2(b) and 2(d)] of each beam was convolved with a line in the X-direction. Figure 3(a) shows a light-sheet obtained with the Bessel-Gauss beam and the corresponding axial cross-section is shown in Fig. 3(d). A large skirt extends over several tens of microns. This intensity distribution is the result of integrating the contributions of all rings along the X-direction. It is worth noting that for shorter Bessel-Gauss beams, the energy within the concentric ring system is reduced, thereby decreasing the magnitude of the skirt. By cleverly choosing the orientation of the sample, Planchon *et al.* used rather short Bessel-Gauss beams to image cells cultured on coverslips and reduce the out-of-focus excitation [11]. Figures 3(b) and 3(c) display light-sheets obtained by extended focusing over 50 and 100 microns, respectively and corresponding cross-sections are shown in Figs. 3(e) and 3(f). Extended focusing also introduces a beam skirt that scales with propagation length, albeit its axial extent is reduced relative to Bessel-Gauss beams.

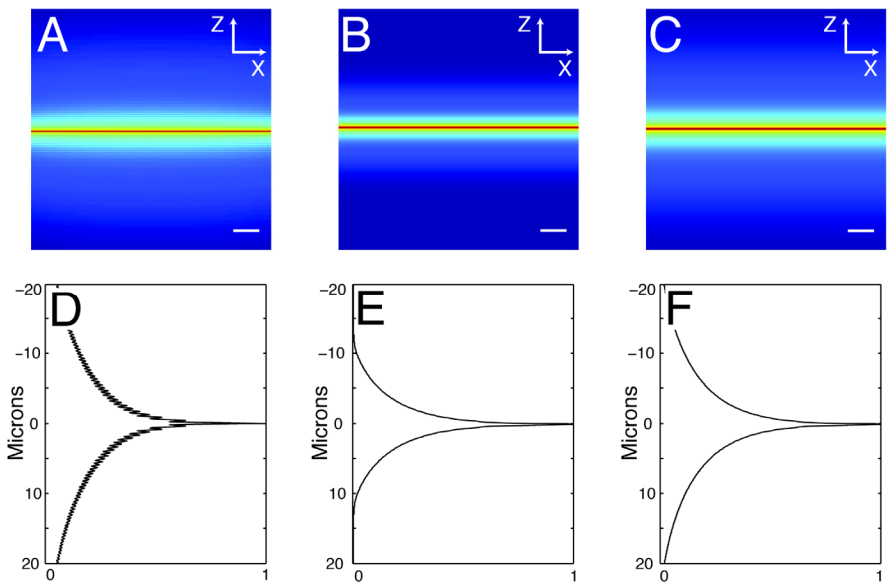

Fig. 3. Simulated axial intensity distributions of light-sheets. Scale bars are 5 microns. (A) Light-sheet generated from a Bessel-Gauss beam. (B) Light-sheet generated from a 50- and (C) 100-micron extended focus Gaussian beam. Profiles of the (D) Bessel-Gauss, (E) 50-micron, and (F) 100-micron extended focus Gaussian beam.

3.2 Transmission measurements of the extended focus

To characterize the extended focus we imaged its beam profile in transmission. For this purpose the whole detection arm was rearranged such that it was collinear with the optical axis of the illumination objective and both objectives shared the same focal plane. With the TAG lens powered off, a shear plate confirmed that the input beam into the illumination objective and the output beam from the detection objective were both well collimated.

We acquired 3D data sets by Z-stepping the detection objective over 100 microns. Figures 4(a) and 4(c) show a maximum intensity projection of a normal laser focus (TAG lens turned off) and an extended focus (TAG lens running at 191kHz and 30% power) as measured in transmission. Corresponding lateral cross sections through the beam are shown in Figs. 4(b) and 4(d). Figures 4(e) and 4(f) show line profiles along the propagation direction and the lateral direction through the two beams, respectively. The lateral FWHM of the Gaussian beam is 497 nm whereas the extended focus measured 726 ± 77 nm (mean and standard

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deviation measured over 80 microns along the Y-direction). Of note, the FWHM drops to 546 nm and 592 nm at each end of the beam. As the laser spot is moving sinusoidally back and forth, the end points of the extended focus are brighter than the central part of the beam [Figs. 4(c) and 4(e)]. For practical imaging, we decided to use the central part of the extended focus to illuminate a 512 x 512 pixel field of view on the camera (corresponding to an area of 82 x 82 microns).

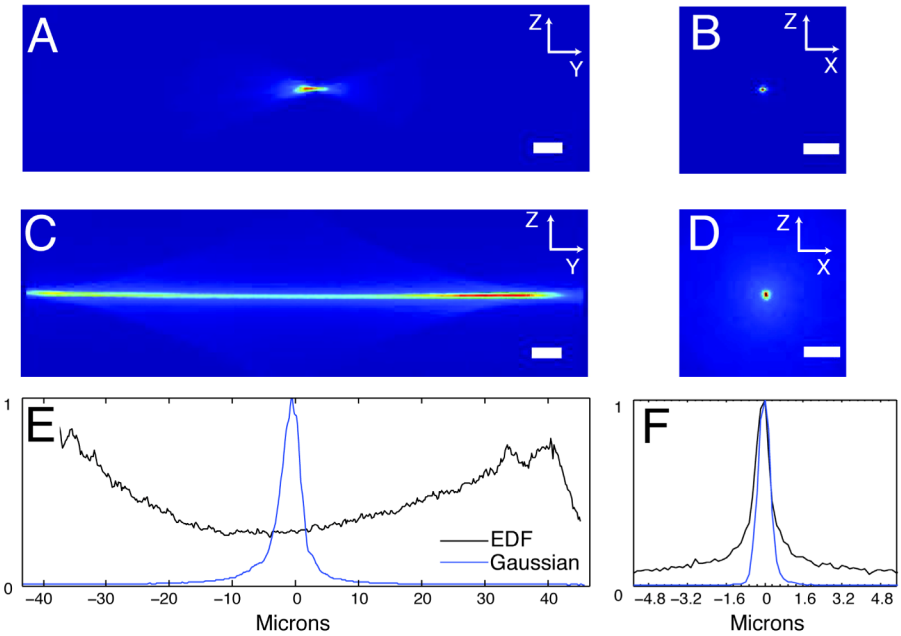

Fig. 4. Measured intensity distributions of Gaussian and extended focus beams. Scale bars are 5 microns. (A) Maximum intensity projection of the propagation of a Gaussian beam. (B) Cross-section of Gaussian beam at laser focus. (C) Maximum intensity projection of the propagation of an extended focus Gaussian beam. (D) Cross-section of extended focus Gaussian beam at middle of the propagation distance. (E) Intensity profile for the extended focus (black) and Gaussian beams (blue) in the propagation direction. (F) Overlay of the beam cross-sections for the extended focus (black) and Gaussian beams (blue) at the center of the propagation trajectory.

3.3 Fluorescent beads imaging

200 nm fluorescent nanospheres embedded in a cube of agarose were used to characterize the point spread function (PSF) and overall imaging performance of the LSFM. An axial step size of 200 nm was used throughout, generating a near-isotropic voxel size (the lateral pixel dimensions are 160 nm).

Figure 5(a) shows a cross-section through an ensemble averaged PSF as obtained with a light-sheet using extended focus illumination and normal orthogonal widefield detection. The PSF features a bright maximum, yet the residual beam skirt results in a slightly elongated PSF in the Z-direction. In contrast to a Bessel-Gauss LSFM, the PSF does not exhibit side lobes, but rather decays monotonically in Z. Figures 5(b) and 5(c) show the PSF obtained by descanning the extended focus as it sweeps the field laterally with a 480 and 160 nm wide (in sample space) virtual confocal slit aperture, respectively. A 480 nm aperture (corresponding to 3 camera pixels) considerably shrinks the PSF in the axial direction. However, using a slit aperture of 160 nm (corresponding to 1 camera pixel) does not noticeably improve the PSF any further. Figure 5(d) shows the PSF after linear Wiener deconvolution [19] of the 480 nm confocal aperture data set. Here, the axial FWHM is 698 nm, which is in agreement with the

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FWHM of the extended focus that was measured in transmission. The lateral FWHM equals to 460 nm.

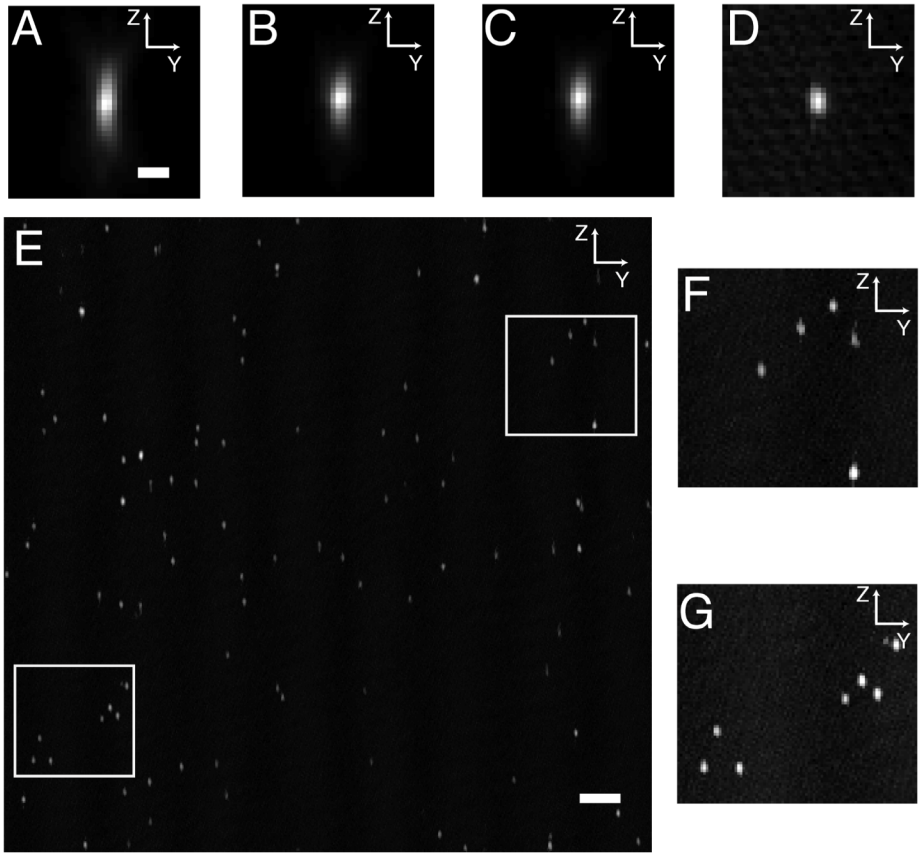

Fig. 5. Extended focus PSF measurements using sub-diffraction fluorescent nanospheres. (A) Single nanosphere in standard camera-based widefield imaging mode. Scale bar is 1 micron. (B) PSF of nanosphere imaged with a 480 nm and (C) 160 nm virtual confocal slit. (D) Linearly deconvolved nanosphere showing near-isotropic axial and lateral resolution. (E) YZ maximum intensity projection through an 80-micron tall Z-stack.

A YZ rendering (maximum intensity projection) of a 3D bead data set using a virtual slit aperture of 480 nm and linear deconvolution with Wiener filtering [19] is shown in Fig. 5(e). The light-sheet microscope maintains shift-invariant imaging throughout the imaging volume (82 x 82 x 80 microns), as evidenced by the enlarged boxed regions shown in Figs. 5(f) & 5(g), which are both located near the edges of the imaging volume.

3.4 Live cell imaging

Live imaging of metastatic melanoma cells was performed at room temperature and the sample chamber of the microscope was filled with phosphate buffered saline. For maximum rejection of out-of-focus blur, confocal descanning was performed with a virtual slit of 160 nm width, the exposure time for one image frame was set to 20 milliseconds, and the axial step size to 200 nm. Each 3D stack encompassed 50 microns axially and took 5.3 s to acquire. An additional pause of 20 s was inserted between time points. Despite the physiologically non-ideal conditions, blebbing (*e.g.*, dynamic hemispherical membrane protrusions) and movement of the cells was observed [20]. Figures 6(a), 6(b), and 6(c) display the maximum intensity projections along the Z, Y and X direction, respectively. Figure 6(d) shows six XY

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cross sections at different depths through a single cell. Individual blebs can be clearly resolved and sectioned.

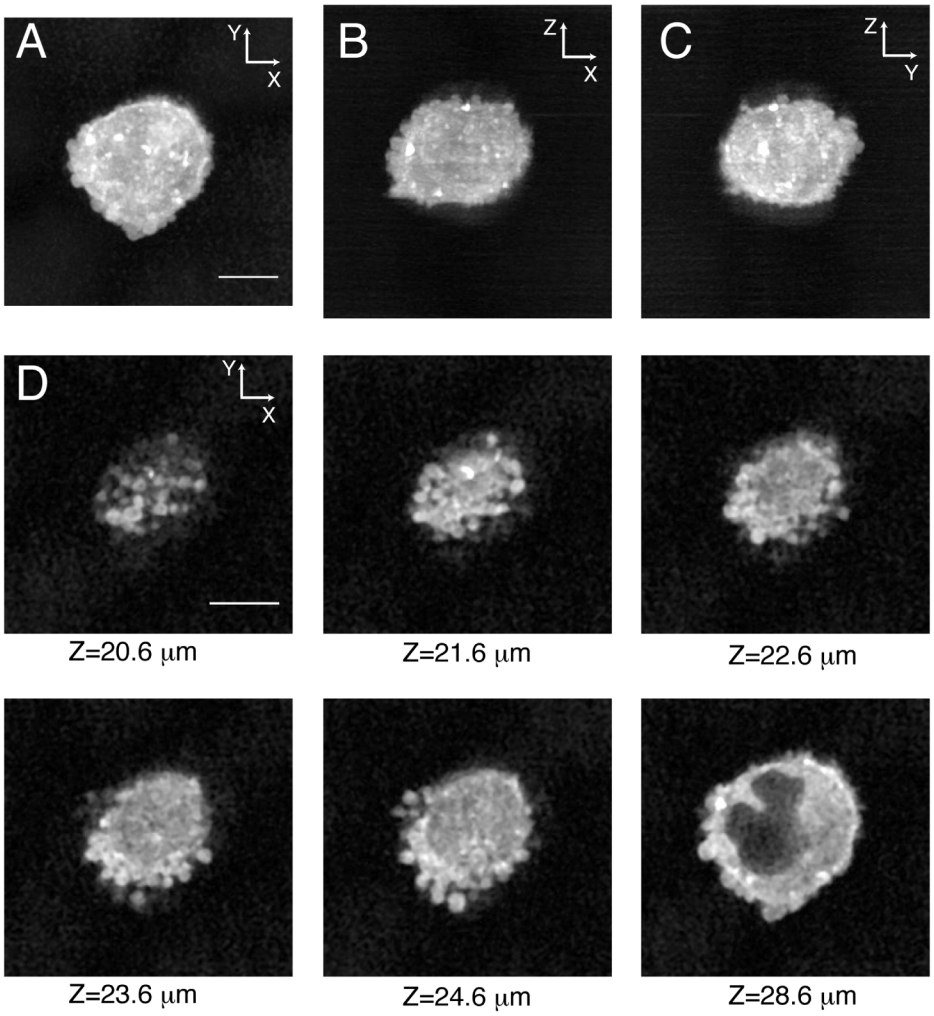

Fig. 6. (A) XY, (B) XZ, and (C) YZ maximum intensity projections of melanoma cell. (D) Individual XY slices through the same melanoma cell. Scale bar is 10 microns.

Figure 7 shows a maximum intensity projection along the optical axis for three time points of a time-lapse series encompassing 10 time points. The arrows mark two large blebs in the first time point. The bleb on the left leaves an F-actin 'scar' after its decay that is visible at t = 102 s. A movie of the whole times series is shown in Media 1.

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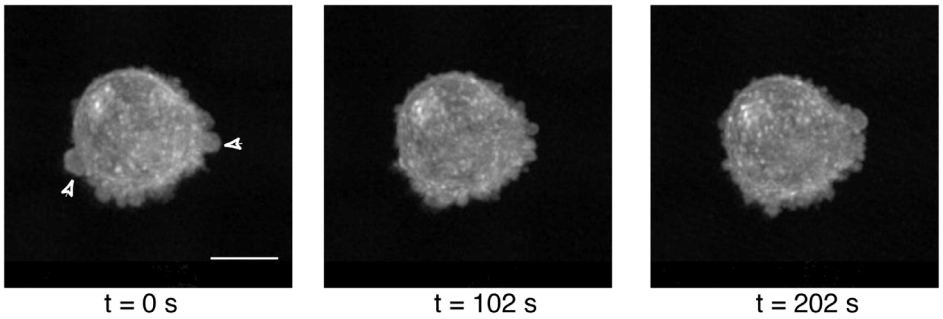

Fig. 7. Time-series of dynamic bleb protrusions in a metastatic melanoma cell line (Media 1). Highlighted regions are shown with white arrowheads. Scale bar is 10 microns.

3.5 Imaging of larger specimens

To demonstrate extended focus LSFM on a larger specimen, we imaged fluorescently labeled collagen. Confocal descanning was performed with a 4 pixel (640 nm in sample space) virtual slit and the image stack was acquired with an axial step size of 300 nm. Figure 8(a) shows a linearly deconvolved maximum intensity projection in XY spanning an image volume of 82x82x80 microns. Figures 8(b) and 8(c) show the raw and deconvolved maximum intensity projections in XZ, respectively, over 40 microns in Y. Residual blur present in the raw data is almost completely removed by linear deconvolution. Individual fibers can be clearly distinguished in the lateral as well as the axial view.

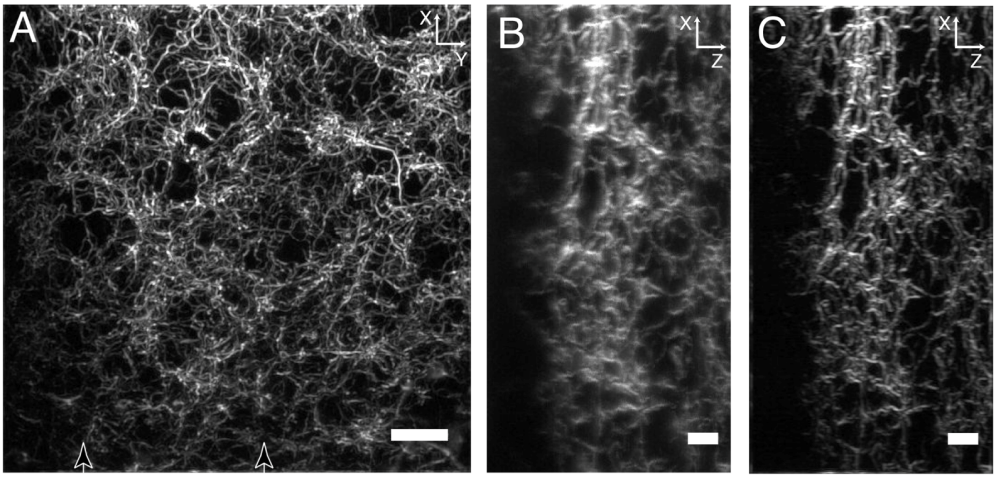

Fig. 8. Imaging of fluorescently labeled collagen. (A) XY maximum intensity projection spanning 80 microns in Z following linear deconvolution. (B) XZ maximum intensity projections spanning 40 microns in Y before and (C) after linear deconvolution. Scale bars in each image are 10 microns. Arrowheads mark the region of (A) represented in (B) and (C).

4. Discussion

We have introduced a new concept for generating light-sheet illumination using extended focusing. Compared to Gaussian and Bessel-Gauss beams, a more uniform lateral illumination field results. Additionally, only the sample volume in the immediate proximity of the field of view is illuminated. In contrast, to achieve a similarly flat illumination field using Bessel-Gauss beams would require a significantly larger propagation length, illuminating cells beyond the field of view and increasing the strength of the side lobes. Even for equivalent propagation distances, our simulations demonstrate that extended focusing results

#221709 - $15.00 USD Received 26 Aug 2014; revised 3 Oct 2014; accepted 3 Oct 2014; published 16 Oct 2014
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in a higher confinement of excitation power to the focal plane than Bessel-Gauss illumination. However, using linear optics (*e.g.,* 1-photon), the loss of axial confinement appears to be a general consequence of extending the depth-of-focus of the beam (*e.g.*, Bessel-Gauss, extended focusing), leading to increased photodamage compared to a LSFM that uses a low-NA Gaussian beam. So far, only 2-photon Bessel-Gauss LSFM can combine large field of views, tight confinement of the excitation power, and high axial resolution.

Light-sheets obtained by extended focusing or Bessel-Gauss beams exhibit a side lobe structure (skirt) that extends beyond the detection objective's depth-of-focus. Consequently, at least for dense samples, out-of-focus blur must be rejected in order to achieve optimal imaging performance. We chose a confocal detection scheme with a virtual slit because it enables faster image acquisition than structured illumination, which requires multiple images per Z-plane. While the confocal aperture can remove the majority of the out-of-focus blur, linear deconvolution is needed to achieve the full axial resolution of the system. For sparse samples, it seems possible to use linear deconvolution alone to remove out-of-focus blur. For densely labeled samples (*e.g.*, the actin cortex), however, descanned confocal detection combined with linear deconvolution became necessary to restore optimal imaging performance. The labeling density at which confocal detection is necessary has not been investigated in this study.

Importantly, as presented in this paper, extended focusing can be adopted by other LSFM modalities with little modification to improve illumination uniformity and increase the lateral field of view. These include LSFMs that implement Gaussian, line-shaped Gaussian, or Bessel-Gauss illumination. Using a focused line instead of a focused spot to generate a light-sheet by extended focusing has three apparent advantages. (A) Only an axial scan by the focus tunable lens is needed to generate a light-sheet. (B) In contrast to the point illumination employed here, the peak power delivered to the sample is decreased substantially as a larger area is simultaneously illuminated. (C) A confocal aperture, virtual or physical, could capture the focused part of the line alone as it is scanned axially (Y-direction in our coordinate definition). Out-of-focus blur, located before and after the focused line, could be completely rejected and diffraction-limited performance could be obtained. It is expected that the out-of-focus rejection would work much more efficiently than in the case of Bessel-Gauss [13] or extended focus beams, as both beams are surrounded by side lobe structures. Thus some amount of blur can slip through the slit aperture, which necessitates excessively small apertures that in turn also clip away valuable in focus light.

Alternative implementations could further improve imaging performance. For example, the resonant nature of the tunable lens creates uneven illumination (*i.e.*, the laser focus spends on average more time at its end points than in the middle) that could be corrected by modulating the drive voltage of the AOM and hence the laser intensity. Furthermore, the sample, located at the middle of the beam trajectory, is unduly illuminated by out-of-focus light stemming primarily from the bright end points. Together, linearizing the laser intensity across the propagation distance of the focus tunable beam would decrease illumination dose, improve image field flatness, and decrease out-of-focus blur. Lastly, an aberration free optical system would further increase the peak of the central lobe, and hence reduce the relative strength of the side lobes.

Even if ideal operation of the TAG lens is achieved, it will only change the defocus of the wavefront. In order to maintain a diffraction-limited focus over a large axial range, spherical and higher wavefront modes need to be actively controlled. Aberration free focusing could be achieved either by objective scanning or by remote focusing methods [21]. Scanning the objective may be too slow for the light-sheet generation presented in this paper, but it might be suitable for scanning a line-focus axially as only one sweep is needed per image frame. Diffraction-limited remote focusing has been demonstrated at rates of 2.7 kHz and up to 200 microns propagation distance with an NA of 0.8 [22], which makes it suitable for either extended focusing method.

In summary we have presented a novel way to generate divergence free light-sheets that exhibit exceptional flatness of illumination. Compared to other methods, highly uniform

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illumination can be achieved with reduced out-of-focus generation. Using the resonant TAG lens, it is conceivable to add the extended focusing capability to existing light-sheet systems. The adoption of LSFM has rapidly accelerated our understanding of developmental biology. With further improvements in sub-cellular LSFM, we anticipate a similar revolution in cellular biology.

Acknowledgments

The Cancer Prevention Research Institute of Texas (CPRIT) funded this research (grant R1225, to Prof. Gaudenz Danuser). The central software employed in this research is used under license from Howard Hughes Medical Institute, Janelia Farm Research Campus. We thank Gerard Garrelts and Dan Milkie of Coleman technologies for their work and support for the LabView control software and Prof. Rainer Heintzmann for fruitful discussions and help on PSF modelling. The authors are also grateful to Prof. Gaudenz Danuser for guidance and support of this research.